

SDP 302

# Problem Set: Monte Carlo Method & OLS

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## Instructions & Information

The **Monte Carlo Method** is an elegant and powerful computational technique that uses repeated sampling to solve complex problems with uncertain variables the solutions of which are difficult or impossible to calculate deterministically. Instead of using rigid analytical methods to find a single, exact deterministic solution to the problem, the Monte Carlo method aims to estimate the outcome of uncertain events by running a plethora of repeated random samplings to provide a range of possible outcomes along with the probability of each occurring. The method goes back to Metropolis and Ulam 1949. Named after the famous casino in Monte Carlo due to its heavy reliance on chance and probability, Monte-Carlo techniques are widely considered a cornerstone in Bayesian statistics and modern data science. Due to their simplicity and versatility they have also become a key method in quantitative research in many scientific fields. The book by Fishman 1996 offers an excellent introduction to Monte Carlo methods, providing both theoretical foundations and practical applications.

In this assignment we will use the Monte Carlo method to demonstrate the following key theoretical properties of the Ordinary Least Squares (OLS) estimator under standard Gauss-Markov assumptions:

1. Unbiasedness:  $\mathbb{E}[\hat{\beta}_i] = \beta_i$ , i.e. on average, the estimated coefficients equal the true population parameters.
2. Consistency and decreasing variance: As sample size  $N$  increases, the distribution of  $\hat{\beta}$  collapses around the true value.
3. Sampling distribution and normality: According to the central limit theorem, the estimators  $\hat{\beta}_i$  follow a normal distribution in large samples even when error terms are non-normal.

To evaluate these properties, we define the following known true population model as our benchmark:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \epsilon_i$$

where true parameters are  $\beta_0 = 1$ ,  $\beta_1 = -3$ ,  $\beta_2 = 2$ . The error terms,  $\epsilon_i$ , are independent and identically distributed, following either a normal distribution,  $\epsilon_i \sim \mathcal{N}(0, 6)$ , or a uniform distribution,  $\epsilon_i \sim \mathcal{U}(-9, +9)$ . This underlying model acts as our Data Generating Process (DGP). We will use it to draw synthetic samples of varying size  $N$  across  $M$  independent simulation runs, allowing us to compare our OLS estimates against the known population parameters.

## References

- Fishman, George S. (1996). *Monte Carlo: Concepts, Algorithms, and Applications*. Springer Series in Operations Research. Springer.
- Metropolis, Nicholas and S. Ulam (1949). "The Monte Carlo Method". In: *Journal of the American Statistical Association* 44.247, pp. 335–341.

**[EXR] Exercise 1 (Exploring the BLUE Properties of OLS)**

In the course *GitHub* repository, you will find two files: `MonteCarlo_BLUE.csv` and `MonteCarlo_BLUE.parquet`. These files contain OLS estimates generated from  $M = 10,000$  simulation runs for four different sample sizes:  $N = 15$ ,  $N = 50$ ,  $N = 200$ , and  $N = 1,000$ . Import either file into a pandas DataFrame to answer the following questions. Note that the final column contains a binary flag indicating the error term distribution which can be used to filter the data.

**Repository**

- a) Explain why OLS linear regression remains valid for estimating  $\beta_0, \beta_1$ , and  $\beta_2$  despite the non-linear term  $X^2$  in the population model.
- b) Filter the dataset for samples where the error term is normally distributed. For each sample size  $N$ :
  - Calculate the mean of the 10,000 estimates for each coefficient and compare these averages to the true population parameters.
  - Plot a histogram showing the distribution of each estimator, clearly marking the true parameter value on each plot.
  - Describe how the centre, spread, and shape of these distributions change as  $N$  increases. Relate your observations directly to the three theoretical properties listed above.
- c) Filter the dataset for samples with uniformly distributed error terms. Repeat the steps in part b). Do your observations or conclusions change? If so, explain what drives these differences.